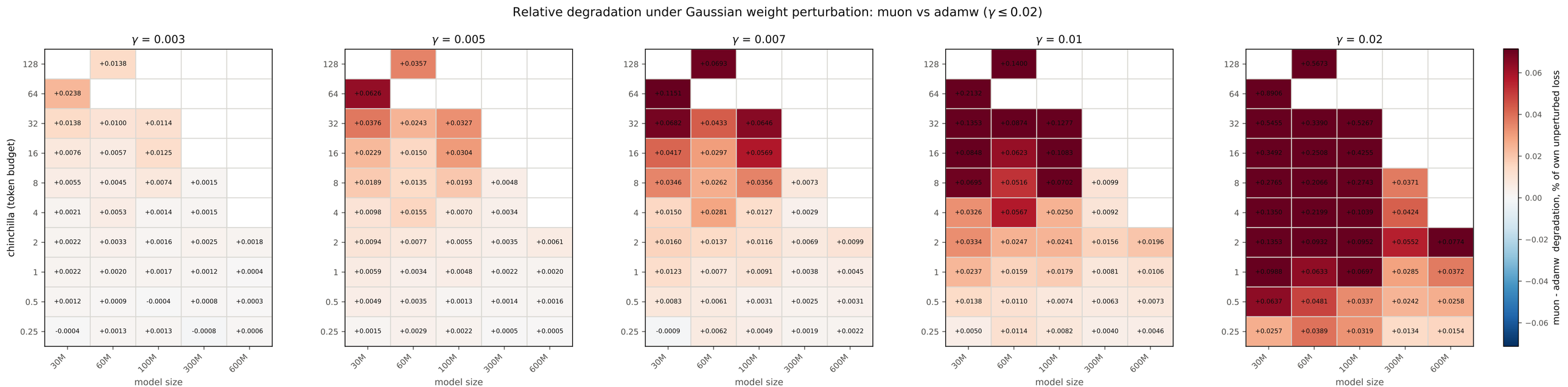
Relative degradation under Gaussian weight perturbation: muon vs adamw ($\gamma \leq 0.02$)
γ = 0.003
γ = 0.005
γ = 0.007
γ = 0.01
γ = 0.02
chinchilla (token budget)
128 64 32 16 8 4 2 1 0.5 0.25
+0.0138 +0.0238 +0.0100 +0.0114 +0.0076 +0.0057 +0.0125 +0.0055 +0.0045 +0.0074 +0.0015 +0.0021 +0.0053 +0.0014 +0.0015 +0.0022 +0.0033 +0.0016 +0.0025 +0.0018 +0.0022 +0.0020 +0.0017 +0.0012 +0.0004 +0.0012 +0.0009 -0.0004 +0.0008 +0.0003 -0.0004 +0.0013 +0.0013 -0.0008 +0.0006
+0.0357 +0.0626 +0.0376 +0.0243 +0.0327 +0.0229 +0.0150 +0.0304 +0.0189 +0.0135 +0.0193 +0.0048 +0.0098 +0.0155 +0.0070 +0.0034 +0.0094 +0.0077 +0.0055 +0.0035 +0.0061 +0.0059 +0.0034 +0.0048 +0.0022 +0.0020 +0.0049 +0.0035 +0.0013 +0.0014 +0.0016 +0.0015 +0.0029 +0.0022 +0.0005 +0.0005
+0.0693 +0.1151 +0.0682 +0.0433 +0.0646 +0.0417 +0.0297 +0.0569 +0.0346 +0.0262 +0.0356 +0.0073 +0.0150 +0.0281 +0.0127 +0.0029 +0.0160 +0.0137 +0.0116 +0.0069 +0.0099 +0.0123 +0.0077 +0.0091 +0.0038 +0.0045 +0.0083 +0.0061 +0.0031 +0.0025 +0.0031 -0.0009 +0.0062 +0.0049 +0.0019 +0.0022
+0.1400 +0.2132 +0.1353 +0.0874 +0.1277 +0.0848 +0.0623 +0.1083 +0.0695 +0.0516 +0.0702 +0.0099 +0.0326 +0.0567 +0.0250 +0.0092 +0.0334 +0.0247 +0.0241 +0.0156 +0.0196 +0.0237 +0.0159 +0.0179 +0.0081 +0.0106 +0.0138 +0.0110 +0.0074 +0.0063 +0.0073 +0.0050 +0.0114 +0.0082 +0.0040 +0.0046
+0.5673 +0.8906 +0.5455 +0.3390 +0.5267 +0.3492 +0.2508 +0.4255 +0.2765 +0.2066 +0.2743 +0.0371 +0.1350 +0.2199 +0.1039 +0.0424 +0.1353 +0.0932 +0.0952 +0.0552 +0.0774 +0.0988 +0.0633 +0.0697 +0.0285 +0.0372 +0.0637 +0.0481 +0.0337 +0.0242 +0.0258 +0.0257 +0.0389 +0.0319 +0.0134 +0.0154
30M 60M 100M 300M 600M
model size
muon - adamw degradation, % of own unperturbed loss
0.06 0.04 0.02 0.00 −0.02 −0.04 −0.06
cell = 100 x (perturbed - unperturbed) / unperturbed for muon minus the same for adamw | negative (blue) = muon degrades less | white = no data
colour scale clipped at ±0.071 (85th pct); 27 cell(s) beyond it keep their printed value